

APPENDIX C

PERMANENT SYMBOL TABLE (PST)

The mnemonics for the PDP-11 operation (op) codes and MACRO-11 assembler directives are stored in the Permanent Symbol Table (PST). The PST contains the symbols that are automatically recognized by MACRO-11.

For a detailed description of the op codes, see the PDP-11 Processor Handbook.

C.1 OP CODES

Instruction Mnemonic	Octal Value	Operation
ADC	005500	Add Carry
ADCB	105500	Add Carry (Byte)
ADD	060000	Add Source To Destination
ASH	072000	Shift Arithmetically
ASHC	073000	Arithmetic Shift Combined
ASL	006300	Arithmetic Shift Left
ASLB	106300	Arithmetic Shift Left (Byte)
ASR	006200	Arithmetic Shift Right
ASRB	106200	Arithmetic Shift Right (Byte)
BCC	103000	Branch If Carry Is Clear
BCS	103400	Branch If Carry Is Set
BEQ	001400	Branch If Equal
BGE	002000	Branch If Greater Than Or Equal
BGT	003000	Branch If Greater Than
BHI	101000	Branch If Higher
BHIS	103000	Branch If Higher Or Same
BIC	040000	Bit Clear
BICB	140000	Bit Clear (Byte)
BIS	050000	Bit Set
BISB	150000	Bit Set (Byte)
BIT	030000	Bit Test
BITB	130000	Bit Test (Byte)
BLE	003400	Branch If Less Than Or Equal
BLO	103400	Branch If Lower
BLOS	101400	Branch If Lower Or Same
BLT	002400	Branch If Less Than
BMI	100400	Branch If Minus
BNE	001000	Branch If Not Equal
BPL	100000	Branch If Plus
BPT	000003	Breakpoint Trap
BR	000400	Branch Unconditional

PERMANENT SYMBOL TABLE (PST)

Instruction Mnemonic	Octal Value	Operation
BVC	102000	Branch If Overflow Is Clear
BVS	102400	Branch If Overflow Is Set
CALL	004700	Jump To Subroutine (JSR PC,xxx)
CALLR	000100	Jump (JMP addr)
CCC	000257	Clear All Condition Codes
CLC	000241	Clear C Condition Code Bit
CLN	000250	Clear N Condition Code Bit
CLR	005000	Clear Destination
CLRB	105000	Clear Destination (Byte)
CLV	000242	Clear V Condition Code Bit
CLZ	000244	Clear Z Condition Code Bit
CMP	020000	Compare Source To Destination
CMPB	120000	Compare Source To Destination (Byte)
COM	005100	Complement Destination
COMB	105100	Complement Destination (Byte)
DEC	005300	Decrement Destination
DECB	105300	Decrement Destination (Byte)
DIV	071000	Divide
EMT	104000	Emulator Trap
FADD	075000	Floating Add
FDIV	075030	Floating Divide
FMUL	075020	Floating Multiply
FSUB	075010	Floating Subtract
HALT	000000	Halt
INC	005200	Increment Destination
INCB	105200	Increment Destination (Byte)
IOT	000004	Input/Output Trap
JMP	000100	Jump
JSR	004000	Jump To Subroutine
MARK	006400	Mark
MED6X	076600	PDP-11/60 Maintenance
MED74C	076601	PDP-11/74 CIS Maintenance
MFPI	006500	Move From Previous Instruction Space
MFPS	106700	Move from PS (LSI-11, LSI-11/23, LSI-11/2)
MFPT	000007	Move From Processor Type
MOV	010000	Move Source To Destination
MOVB	110000	Move Source To Destination (Byte)
MTPI	006600	Move To Previous Instruction Space
MTPS	106400	Move to PS (LSI-11, LSI-11/23, LSI-11/2)
MUL	070000	Multiply
NEG	005400	Negate Destination
NEGB	105400	Negate Destination (Byte)
NOP	000240	No Operation
RESET	000005	Reset External Bus
RETURN	000207	Return From Subroutine (RTS PC)
ROL	006100	Rotate Left
ROLB	106100	Rotate Left (Byte)
ROR	006000	Rotate Right
RORB	106000	Rotate Right (Byte)

PERMANENT SYMBOL TABLE (PST)

Instruction Mnemonic	Octal Value	Operation
RTI	000002	Return From Interrupt (Permits a trace trap)
RTS	000200	Return From Subroutine
RTT	000006	Return From Interrupt (inhibits trace trap)
SBC	005600	Subtract Carry
SBCB	105600	Subtract Carry (Byte)
SCC	000277	Set All Condition Code Bits
SEC	000261	Set C Condition Code Bit
SEN	000270	Set N Condition Code Bit
SEV	000262	Set V Condition Code Bit
SEZ	000264	Set Z Condition Code Bit
SOB	077000	Subtract One And Branch
SUB	160000	Subtract Source From Destination
SWAB	000300	Swap Bytes
SXT	006700	Sign Extend
TRAP	104400	Trap
TST	005700	Test Destination
TSTB	105700	Test Destination (Byte)
TSTSET	007200	Test Destination And Set Low Bit
WAIT	000001	Wait For Interrupt
WRTLCK	007300	Read/Lock Destination. Write/Unlock R0 Into Destination
XFC	076700	Extended Function Code
XOR	074000	Exclusive OR

COMMERCIAL INSTRUCTION SET (CIS) OP CODES

Every operation listed in the CIS table has two instruction mnemonics. The suffix "I", attached to every second mnemonic, indicates that the addresses are inline. The inline instructions require two arguments; the other instructions (excepting L2DN and L3DN) require no arguments.

Instruction Mnemonic	Octal Value	Operation
ADDN	076050	Add Numeric
ADDNI	076150	Add Numeric
ADDP	076070	Add Packed
ADDPI	076170	Add Packed
ASHN	076056	Arithmetic Shift Numeric
ASHNI	076156	Arithmetic Shift Numeric
ASHP	076076	Arithmetic Shift Packed
ASHPI	076176	Arithmetic Shift Packed
CMPC	076044	Compare Character String
CMPCI	076144	Compare Character String
CMPN	076052	Compare Numeric
CMJNI	076152	Compare Numeric
CMPP	076072	Compare Packed
CMPI	076172	Compare Packed
CVTLN	076057	Convert Long To Numeric
CVTLNI	076157	Convert Long To Numeric
CVTLN	076077	Convert Long To Packed

PERMANENT SYMBOL TABLE (PST)

Instruction Mnemonic	Octal Value	Operation
CVTLPI	076177	Convert Long To Packed
CVTNP	076055	Convert Numeric To Packed
CVTNPI	076155	Convert Numeric To Packed
CVTPN	076054	Convert Packed To Numeric
CVTPNI	076154	Convert packed To Numeric
DIVP	076075	Divide Decimal
DIVPI	076175	Divide Decimal
LOCC	076040	Locate Character
LOCCI	076140	Locate Character
L2DN*	07602N	Load 2 Descriptors @(RN)+
L3DN*	07606N	Load 3 Descriptors @(RN)+
MATC	076045	Match Character
MATCI	076145	Match Character
MOVC	076030	Move Character
MOVCI	076130	Move Character
MOVRC	076031	Move Reverse Justified Character
MOVRCI	076131	Move Reverse Justified Character
MOVTC	076032	Move Translated Character
MOVTCI	076132	Move Translated Character
MULP	076074	Multiply Decimal
MULPI	076174	Multiply Decimal
SCANC	076042	Scan Character
SCANCI	076142	Scan Character
SKPC	076041	Skip Character
SKPCI	076141	Skip Character
SPANC	076043	Span Character
SPANCI	076143	Span Character
SUBN	076051	Subtract Numeric
SUBNI	076151	Subtract Numeric
SUBP	076071	Subtract Packed
SUBPI	076171	Subtract Packed

* where N=0...7

FLOATING POINT PROCESSOR OP CODES

Instruction Mnemonic	Octal Value	Operation
ABSD	170600	Make Absolute Double
ABSF	170600	Make Absolute Floating
ADDD	172000	Add Double
ADDF	172000	Add Floating
CFCC	170000	Copy Floating Condition Codes
CLRD	170400	Clear Double
CLRF	170400	Clear Floating
CMPD	173400	Compare Double
CMFP	173400	Compare Floating
DIVD	174400	Divide Double
DIVF	174400	Divide Floating
LDCDF	177400	Load And Convert From Double To Floating
LDCFD	177400	Load And Convert From Floating To Double
LDCID	177000	Load And Convert Integer To Double

PERMANENT SYMBOL TABLE (PST)

Instruction Mnemonic	Octal Value	Operation
LDCIF	177000	Load And Convert Integer To Floating
LDCLD	177000	Load And Convert Long Integer To Double
LDCLF	177000	Load And Convert Long Integer To Floating
LDD	172400	Load Double
LDEXP	176400	Load Exponent
LDF	172400	Load Floating
LDFPS	170100	Load FPPs Program Status
MFPD	106500	Move From Previous Data Space
MODD	171400	Multiply And Integerize Double
MODF	171400	Multiply And Integerize Floating
MTPD	106600	Move To Previous Data Space
MULD	171000	Multiply Double
MULF	171000	Multiply Floating
NEGD	170700	Negate Double
NEGF	170700	Negate Floating
SETD	170011	Set Double Mode
SETF	170001	Set Floating Mode
SETI	170002	Set Integer Mode
SETL	170012	Set Long Integer Mode
SPL	000230	Set Priority Level
STA0	170005	Diagnostic Floating Point
STB0	170006	Diagnostic Floating Point
STCDF	176000	Store And Convert From Double To Floating
STCDI	175400	Store And Convert From Double To Integer
STCDL	175400	Store And Convert From Double To Long Integer
STCFD	176000	Store And Convert From Floating To Double
STCFI	175400	Store And Convert From Floating To Integer
STCFL	175400	Store And Convert From Floating To Long Integer
STD	174000	Store Double
STEXP	175000	Store Exponent
STF	174000	Store Floating
STFPS	170200	Store FPPs Program Status
STST	170300	Store FPPs Status
SUBD	173000	Subtract Double
SUBF	173000	Subtract Floating
TSTD	170500	Test Double
TSTF	170500	Test Floating

PERMANENT SYMBOL TABLE (PST)

C.2 MACRO-11 DIRECTIVES

The MACRO-11 directives that follow are described in greater detail in Appendix B.

Directive	Function
.ASCII	Translates character string to ASCII equivalents.
.ASCIZ	Translates character string to ASCII equivalents; inserts zero byte as last character.
.ASECT	Begins absolute program section (provided for compatibility with other PDP-11 assemblers).
.BLKB	Reserves byte block in accordance with value of specified argument.
.BLKW	Reserves word block in accordance with value of specified argument.
.BYTE	Generates successive byte data in accordance with specified arguments.
.CROSS	Enables cross-reference listing for specified symbols; enables cross-reference for all symbols.
.CSECT	Begins relocatable program section (provided for compatibility with other PDP-11 assemblers).
.DSABL	Disables specified function.
.ENABL	Enables specified function.
.END	Defines logical end of source program.
.ENDC	Defines end of conditional assembly block.
.ENDM	Defines end of macro definition, repeat block, or indefinite repeat block.
.ENDR	Defines end of current repeat block (provided for compatibility with other PDP-11 assemblers).
.ERROR	Outputs diagnostic message to listing file or command output device.
.EVEN	Word-aligns the current location counter.
.FLT2	Causes two words of storage to be generated for each floating-point argument.
.FLT4	Causes four words of storage to be generated for each floating-point argument.
.GLOBL	Declares global attribute for specified symbol(s).
.IDENT	Labels object module with specified program version number.
.IF	Begins conditional assembly block.
.IFF	Begins subconditional assembly block (if conditional assembly block test is false).
.IFT	Begins subconditional assembly block (if conditional assembly block test is true).
.IFTF	Begins subconditional assembly block (whether conditional assembly block test is true or false).
.IIF	Assembles immediate conditional assembly statement (if specified condition is satisfied).
.INCLUDE	Inserts specified source file within source file currently being used.
.IRP	Begins indefinite repeat block; replaces specified symbol with specified successive real arguments.
.IRPC	Begins indefinite repeat block; replaces specified symbol with value of successive characters in specified string.
.LIBRARY	Adds a specified file name to a macro library list that is searched.

PERMANENT SYMBOL TABLE (PST)

Directive	Function
.LIMIT	Reserves two words of storage for high and low addresses of task image.
.LIST	Controls listing level count and format of assembly listing. .MACRO denotes start of macro definition.
.MCALL	Identifies required macro definition(s) for assembly.
.MDELETE	Deletes the definitions of the specified macro(s).
.MEXIT	Exit from current macro definition or indefinite repeat block.
.NARG	Equates specified symbol to the number of non-keyword arguments in the macro expansion.
.NCHR	Equates specified symbol to the number of characters in the specified character string.
.NLIST	Controls listing level count and suppresses specified portions of the assembly listing.
.NOCROSS	Disables cross-reference listing for specified symbols; disables cross-reference listing for all symbols.
.NTYPE	Equates specified symbols to the addressing mode of the specified argument.
.ODD	Byte-aligns the current location counter.
.PACKED	Generates packed decimal data, 2 digits per byte.
.PAGE	Advances form to top of next page.
.PRINT	Prints specified message on command output device.
.PSECT	Begins specified program section having specified attributes.
.RADIX	Changes current program radix to specified radix.
.RAD50	Generates data block having Radix-50 equivalents of specified character string.
.REM	Inserts a block of comments into a MACRO-11 source program without having to precede comments lines with the comment character (;).
.REPT	Begins repeat block and replicates it according to the value of the specified expression.
.RESTORE	Stores the current program section context on the top of the program section context stack.
.SAVE	Retrieves the program section from the top of the program section context stack.
.SBTTL	Prints specified subtitle text as the second line of the assembly listing page header.
.TITLE	Prints specified title text as object module name in the first line of the assembly listing page header.
.WEAK	Specifies symbols that are either defined externally in another module or are defined globally in the current module.
.WORD	Generates successive word data in accordance with specified arguments.